

28.05.2026

Applied Machine Learning

WORKING WITH IMAGES USING TORCHVISION

- **Quiz**
- **Assignment**
- **Projects**
- **Input**
 - **Key CNN Concepts**
 - **Transfer Learning**
 - **Adversarial Attacks**

QUIZ



<https://forms.office.com/e/HP7B84D4sD>

ASSIGNMENT PRESENTATION

ASSIGNMENT: WHO PRESENTS NEXT?

PROJECTS

PROJECT MILESTONES

- 23.04. Present your Ideas
- 30.04. Form Groups & Put your Name in the [Excel](#)
- 07.05. Literature Review
- 14.05. Dataset Characteristics and Individual Feedback Sessions (**PUBLIC HOLIDAY**)

Deadline for completing the repo sections: 17.05.

- 21.05. Definition of Model Evaluation
- 28.05. Project Work
- 04.06. Baseline Model Estimation

Deadline for completing the repo sections: 07.06.

- 11.06. Project Work / Optional Feedback Sessions (**NO REGULAR CLASS**)
- 25.06. Project Presentations, Part I
- 02.07. Project Presentations, Part II

Submission deadline for the documented repo: 01.08.2026

PROJECTS: BASELINE MODEL

- How good is your final model actually? Metrics like RMSE or accuracy are difficult to interpret by themselves.
- A baseline model is a simple model that provides a context for the performance of your metric.
 - In regression problems, this might be a linear model.
 - In time series problems, this might be a simple forecast of the last known value.
- It also requires you to think about metrics, data processing, train-test splits, which you need for the complex model as well.

KEY CNN CONCEPTS (IN COMPUTER VISION)

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- **Kernel:** a matrix of weights applied across the input image (slid over the image: only ever applied to a small part of the image at a time)
- **Stride:** number of pixels by which the kernel moves across the image
- **Padding:** extra pixels (usually zeros, i.e., black) added around the image so the kernel can slide over the edges of the image
- **Feature map:** 2D output from applying one filter over the image
- **Pooling:** reduces information from multiple pixels to a single value

TRANSFER LEARNING

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- Use the **correct preprocessing** for the backbone model
 - Matching image size
 - Matching scaling of pixel values (e.g., ImageNet values)
- **Start simple, iterate fast**
 - Small backbone (e.g., ResNet-18, MobileNetV3)
 - Train only the head
 - Iteratively increase complexity: larger backbones, unfreezing more layers
- See the [TorchVision documentation](#) for an overview and code snippets

ADVERSARIAL ATTACKS

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- Neural Networks do not „see“ the same way that humans do
- See adversarial attacks ([Shi et al., 2020](#)):

